Most-cited prototypes. Units: L<layer>u<unit>; ¬ = must be 0; layer-0 units decoded into atom-count rules (n_X = number of X among the node and its neighbours)

P51 cited by 80 test explanations (class 1 50, class 0 30)

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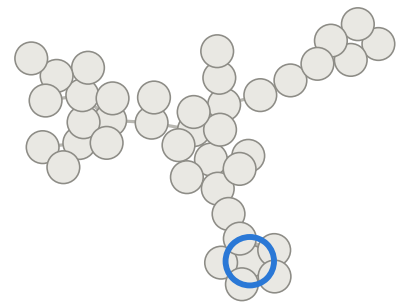
rule:  L0u1 ∧ L0u18 ∧ L0u55 ∧ L0u61 ∧ ¬L2u5 ∧ ¬L2u33 ∧ ¬L2u37 ∧ ¬L2u46 ∧ ¬L2u48 ∧ ¬L2u51 ∧ L2u54
training graphs with a fully matching node: class 0 34%, class 1 100%
cited as: mean over nodes of match(P51) ≤ 0.34 or ≥ 0.87  (x50)
cited as: no node matches P51  (x30)
   L0u1 = n_Br in [0.30, 2.50]
   L0u18 = (n_Br in [0.30, 2.50]) ∨ (n_N in [0.43, 2.50])
   L0u55 = (n_N in [0.43, 2.50] ∧ n_Br in [0.30, 2.50] ∧ n_NOT P in [2.42, 3.81] U [5.16, 6.57] U [7.91, 9.30] U [10.65,
       12.06] U [13.40, 14.79] U [16.16, 17.55] U [18.89, 20.28] U [21.65, 22.50]) ∨ (n_N in [0.43, 2.50] ∧ n_NOT H in [1.80,
       1.90] U [3.85, 5.86] U [7.81, 9.82] U [11.77, 13.78] U [15.72, 17.73] U [19.68, 21.69] ∧ n_NOT P in [2.42, 3.81] U
       [5.16, 6.57] U [7.91, 9.30] U [10.65, 12.06] U [13.40, 14.79] U [16.16, 17.55] U [18.89, 20.28] U [21.65, 22.50])
   L0u61 = (n_NOT H in [1.80, 1.90] U [3.85, 5.86] U [7.81, 9.82] U [11.77, 13.78] U [15.72, 17.73] U [19.68, 21.69] ∧ n_NOT
       N in [1.80, 4.35] U [8.41, 12.49] U [16.55, 20.64] ∧ n_NOT P in [2.42, 3.81] U [5.16, 6.57] U [7.91, 9.30] U [10.65,
       12.06] U [13.40, 14.79] U [16.16, 17.55] U [18.89, 20.28] U [21.65, 22.50]) ∨ (n_N in [0.43, 2.50])
```

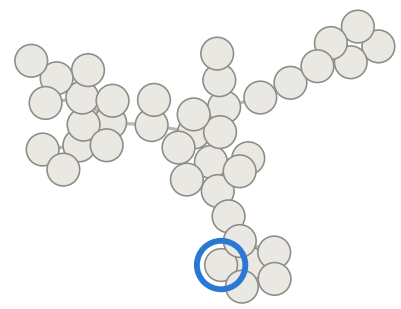
a best-matching training node (graph 4, class 1)



P2 cited by 51 test explanations (class 1 51)

```
rule:  ¬L2u5 ∧ ¬L2u17 ∧ ¬L2u21 ∧ L2u30 ∧ ¬L2u40 ∧ L2u47 ∧ ¬L2u48 ∧ L2u61
training graphs with a fully matching node: class 0 41%, class 1 100%
cited as: mean over nodes of match(P2) in [0.16, 0.49] ∪ [0.82, 1.00]  (x51)
```

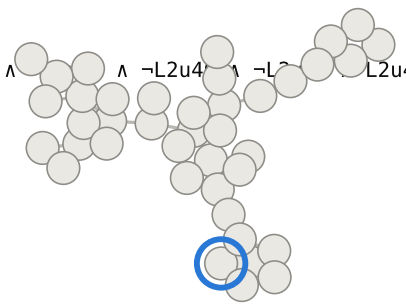
a best-matching training node (graph 4, class 1)



P7 cited by 51 test explanations (class 1 51)

```
rule:  ¬L0u17 ∧ ¬L0u39 ∧ ¬L0u47 ∧ L0u51 ∧ ¬L1u11 ∧ ¬L1u22 ∧ L1u33 ∧ ¬L1u40 ∧ ¬L1u53 ∧ ¬L2u1 ∧ ¬L2u2 ∧ L2u4 ∧ ¬L2u5 ∧ L2u10 ∧ L2u14 ∧ ¬L2u17 ∧ L2u30 ∧ [illegible] ∧ ¬L2u47 ∧ ¬L2u48
∧ ¬L2u49 ∧ L2u54 ∧ L2u58 ∧ L2u61
training graphs with a fully matching node: class 0 41%, class 1 100%
cited as: mean over nodes of match(P7) in [0.07, 0.38] ∪ [0.68, 0.99]  (x51)
   L0u17 = (n_NOT C in [1.80, 2.44] U [5.30, 8.18] U [11.04, 13.94] U [16.80, 19.68] ∧ n_NOT N in [1.80, 4.35] U [8.41,
       12.49] U [16.55, 20.64]) ∨ (n_NOT N in [1.80, 4.35] U [8.41, 12.49] U [16.55, 20.64] ∧ n_NOT S in [1.80, 2.46] U
       [5.30, 8.16] U [11.00, 13.86] U [16.68, 19.56] U [22.38, 22.50])
   L0u39 = no short rule
   L0u47 = n_NOT C in [1.80, 2.44] U [5.30, 8.18] U [11.04, 13.94] U [16.80, 19.68] ∧ n_NOT H in [1.80, 1.90] U [3.85, 5.86]
       U [7.81, 9.82] U [11.77, 13.78] U [15.72, 17.73] U [19.68, 21.69] ∧ n_NOT N in [1.80, 4.35] U [8.41, 12.49] U [16.55,
       20.64]
   L0u51 = (n_S in [0.29, 2.50]) ∨ (n_NOT O in [2.38, 3.77] U [5.14, 6.52] U [7.89, 9.28] U [10.65, 12.04] U [13.40, 14.79] U
       [16.14, 17.55] U [18.89, 20.30] U [21.65, 22.50])
```

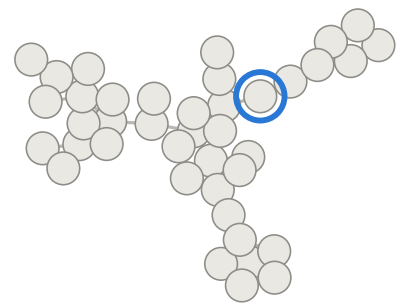
a best-matching training node (graph 4, class 1)



P14 cited by 51 test explanations (class 1 51)

```
rule:  L0u7 ∧ L0u9 ∧ L0u32 ∧ L0u51 ∧ ¬L2u0 ∧ ¬L2u1 ∧ ¬L2u5 ∧ ¬L2u17 ∧ L2u30 ∧ ¬L2u42 ∧ ¬L2u46 ∧ ¬L2u48
training graphs with a fully matching node: class 0 44%, class 1 100%
cited as: mean over nodes of match(P14) in [0.12, 0.49] ∪ [0.87, 1.00]  (x51)
   L0u7 = (n_Br in [0.30, 2.50]) ∨ (n_S in [0.29, 2.50])
   L0u9 = n_S in [0.29, 2.50]
   L0u32 = (n_S in [0.29, 2.50]) ∨ (n_N in [0.43, 2.50])
   L0u51 = (n_S in [0.29, 2.50]) ∨ (n_NOT O in [2.38, 3.77] U [5.14, 6.52] U [7.89, 9.28] U [10.65, 12.04] U [13.40, 14.79] U
       [16.14, 17.55] U [18.89, 20.30] U [21.65, 22.50])
```

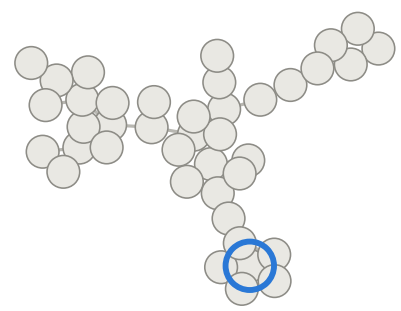
a best-matching training node (graph 4, class 1)



P61 cited by 51 test explanations (class 1 51)

```
rule:  L0u1 ∧ L0u7 ∧ ¬L2u5 ∧ ¬L2u37 ∧ ¬L2u42 ∧ ¬L2u48 ∧ ¬L2u52 ∧ L2u54
training graphs with a fully matching node: class 0 41%, class 1 100%
cited as: mean over nodes of match(P61) ≤ 0.45  (x51)
   L0u1 = n_Br in [0.30, 2.50]
   L0u7 = (n_Br in [0.30, 2.50]) ∨ (n_S in [0.29, 2.50])
```

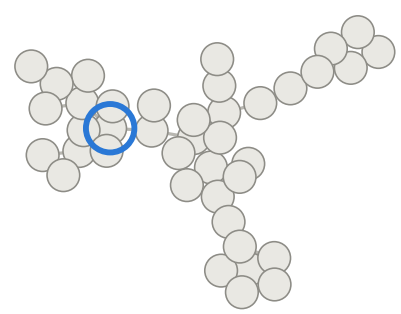
a best-matching training node (graph 4, class 1)



P47 cited by 49 test explanations (class 1 49)

```
rule:  L1u7 ∧ L1u37 ∧ ¬L1u49 ∧ L2u1 ∧ ¬L2u16 ∧ L2u46 ∧ L2u48 ∧ L2u51 ∧ L2u52
training graphs with a fully matching node: class 0 99%, class 1 99%
cited as: mean over nodes of match(P47) ≤ 0.42  (x49)
```

a best-matching training node (graph 4, class 1)



PRELIMINARY